Saved five-loop zero-momentum coefficients

Inspect the completed reduction and its scalar coefficients without repeating IBPs.

55	58	449
individual master identities	formal master products	real Laurent coefficients

The saved expression is the complete zero-momentum contraction of the four-rung quark-loop fixture, with its native factors and unaveraged metric/color contraction. Its 55 identities comprise 46 five-loop masters and nine lower-loop masters: one at 1L, one at 2L, two at 3L and five at 4L. All remain formal, including repeated factors and products. No five-loop master value is computed.

What is expanded

Write the unit-mass result as $g^{10}\sum_\alpha P_\alpha(\varepsilon)C_\alpha(\varepsilon)$, where g is the unchanged `UFO::GC_11` and P_α is a canonical master product with its full multiplicities. The supplied scalar jets obey

$$C_\alpha(\varepsilon)=\sum_{k=v_\alpha}^{5} c_{\alpha,k}\varepsilon^k+\mathcal{O}(\varepsilon^6).$$

The original output already contains $d=4-2\varepsilon$ and the normalization $C_\alpha(\varepsilon)=R_\alpha(\varepsilon)\exp(-5\gamma_{\mathrm{E}}\varepsilon)$. Neither is applied again. Only the derived coefficient artifacts set $M_{\mathrm{UV}}=\mu_{\mathrm{r}}=1$; the scale symbol is absent in this saved input and no scale factor is inserted. The original retained artifact keeps its mass dependence and normalization unchanged.

The remainder is coefficientwise. $\mathcal{O}(\varepsilon^6)$ describes each scalar C_α, with the masters left unexpanded. The masters themselves depend on ε and can have poles, so this is not a fully evaluated physical amplitude through ε^5.

Load or reproduce

Use the existing coherent Symbolica Community installation containing Vakint:

```
python crates/vakint/examples/five_loop_q0_coefficients.py verify
python crates/vakint/examples/five_loop_q0_coefficients.py regenerate \
  --input PATH/result.symbolica --output NEW
```

Verification reads the saved artifacts and their precision/provenance records. Regeneration starts from the retained expression, groups full master products, and evaluates only their scalar jets. Both workflows avoid graph preparation and IBP reduction. Regeneration requires a new output directory.

Artifacts: `examples/data/five_loop_q0/` relative to the Vakint crate. The companion uses ordinary public APIs in one Symbolica kernel, with no private shared-library path, implicit download, or Mathematica requirement.

Precision and faithful interchange

Each combined scalar Laurent coefficient receives one tracked evaluation with 300 requested decimal digits and the fixed 32-digit guard allowance. There is no automatic retry. The minimum measured nonzero precision in this artifact is 330 decimal digits; final rounding is checked separately.

Component	Stored precision	Interpretation
Nonzero real coefficient	998 binary bits	At least 300 decimal digits after export rounding
Exact imaginary zero	1104 binary bits	Exact zero; storage width is not a finite accuracy limit

The sidecar retains componentwise achieved precision, exact rational centers and absolute error radii, error quality and warnings. Radii include final binary rounding. An independent development oracle at 380 digits checked all 449 real coefficients against those saved radii; it is not an end-user dependency.

Artifact	Role
`q0-retained.atom / .expr`	Original complete symbolic reduction
`q0-coefficients-exact-…`	Exact scalar jets through ε^5, masters formal
`q0-coefficients-300digits-…atom`	Checked native Float values with component storage precision
`coefficient-table.json`	Lossless exact-ratio/explicit-bit reconstruction plus uncertainty metadata
`…300digits-…expr`	Human decimal display; not a lossless Float round trip
`provenance.json`	Checksums, source bindings, validation and preserved failure history

The native Atom and exact-ratio JSON reconstruct identically. Canonical decimal printing can lose a binary bit when parsed again; numerical `.expr` files therefore remain displays. The original exact text and exact symbolic jets have their separate checks.

Scientific and postprocessing provenance

Original reduction: executable `490fd6492677`, external data `d4384149b507`, retained expression `da909dc0b8f1`. **Coefficient postprocessor:** actual runtime source `4111049372c0`, host executable `67937cb5291d`. The literal production-source comparison to `03e80d538c6e` covers 676 compiled production inputs; it does not relabel the executable as a rebuild of that later commit. Full identities are recorded in the provenance sidecar.

The 58 exact jets and tracked passes completed in **3.775 s**. A strict export check then exposed a storage mismatch: native exact-zero imaginary components kept 1104 bits, while JSON had recorded 998. A **2.205 s** metadata-only completion repaired those 449 exact-zero precision records and performed the independent validation. The failed receipt remains preserved. Every native and expression file stayed byte-for-byte unchanged; no coefficient was reevaluated.

These are development *postprocessing* observations, not IBP reduction benchmarks or cache speedups. The read-only repair codec is pinned specifically to Symbolica `9cd57678d8817ec33c05cacba4925162f1c86b0b` it makes no claim about future native formats.